Cyrille E. Mvomo

McGill University
Montreal, QC, H3A 0G4, Canada
https://cyrillemvomo.github.io/ | cyrille.mvomo@mail.mcgill.ca | +1 (438) 470 9643

Research Interests

My research is on **"interpretables,"** intelligent wearable sensing systems that reveal age- and disease-related deviations in a person's neurobehavioral state and provide clinicians with information they can use to guide clinical decisions. As a clinician-informatician, I specifically focus on **(1) person-specific modeling of neuromotor deficits** to understand the unique presentation of mobility impairments in each individual (e.g., via ordinal trends canonical variates analysis); **(2) neurobehavioral fingerprint development** to identify signatures inferred from wearables that are associated with the modeled neuromotor deficits (e.g., lyapunov exponents as proxy for altered motor automaticity); and **(3) clinical decision support system design** to translate these neurobehavioral fingerprints into technologies that clinicians can trust to personalize support (e.g., monitoring biomarkers).

Education

2027 (Expected) | **Ph.D.** in Biomechanics and Neurosciences — McGill University, *Montreal, Canada*
Dissertation: *Simple Walking Reveals Less, Complexity Provides Insight: Advancing Digital Gait Biomarkers in Parkinson's Disease*
Committee: Caroline Paquette (chair), Chris Awai and Phil Dixon
cGPA: 4.0/4.0 (*coursework completed*)

2023 | **Visiting Student** in Bioengineering – Robotics — EPFL, *Lausanne, Switzerland*
Advisors: Mohamed Bouri and Alireza Manzoori
Project GPA: 4.0/4.0 [*certificate*]

2023 | **M.Sc.** in Informatics (Digital Sciences) — University of Paris – Cité, *Paris, France*
Concentration in Biomedical Data Science
cGPA: 4.0/4.0 [*certificate*]

2021 | **Bachelor of Podiatric Medicine** (*Clinical Degree*) — University of Paris – Cité, *Paris, France*
B.Sc. Grade in Allied Health Sciences (*University Title*)
cGPA: 3.7/4.0 [*certificate*]

Honors and Awards

24 recognitions for excellence in research, teaching, and service, spanning institutional to international levels.

Fellowships and Scholarships *(all competitive awards independently secured)*

2026	AgeTech Fellowship Top-Up Award, Perley Health (CA$2,500)	*Institutional Level*
2026	Graduate Student Award, Parkinson Canada (CA$40,000)	*Federal level*
2026	Canada Graduate Research Scholarship (CGRS) – Doctoral, Canadian Institutes of Health Research (CIHR) (winning application withdrawn)	*Federal level*
2026	Doctoral Research Scholarship, Quebec Research Fund (CA$58,333)	*Provincial level*
2024	Recruitment Fellowship, Quebec BioImaging Network (CA$10,000)	*Provincial level*
2023	Graduate Excellence Fellowship, McGill University (CA$60,000)	*Institutional level*

2023	Anti-Black Racism Excellence Award, McGill University (CA$10,000)	*Non-competitive*
2023	SMARTS-UP Scholarship, University of Paris – Cité (~CA$7,800)	*Institutional level*
2022	Visiting Scholarship, Lake Lucerne Institute (~CA$8,700)	*Non-competitive*

Best Papers and Presentations

2025	IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI) 2025 – 3rd Prize Best Paper Award (CA$207) [**500+ submissions, Top 0.6%**]	*International level*
2024	Student Colloquium of the Montreal Centre for Interdisciplinary Research in Rehabilitation (CRIR) – People's Choice Award (CA$50)	*Institutional level*

Teaching and Service

2025	Quacquarelli Symonds (QS) – Reimagine Education Bronze Award [**Awarded as a member of the McGill SciLearn team**]	*International level*
2025	IEEE-EMBS BHI 2025 – Reviewer Recognition	*International level*

Travel Awards

2026	Catherine Berg Student Travel Award, CRIR (CA$500)	*Non-competitive*
2025	Travel Award, McGill University – Centre for Research on Brain, Language and Music (CRBLM) (CA$700)	*Institutional level*
2025	Travel Award, Quebec Association for Physical Activity (CA$250)	*Provincial level*
2024	Institute Community Support Travel Award, CIHR (CA$1,700)	*Federal level*
2024	Travel Award, Parkinson Canada (CA$1,000)	*Federal level*

Other Honors and Awards

2025	National Science Foundation–IEEE EMBS–Google, Young Professional NextGen Scholar Recognition (~CA$415)	*International level*
2025	McGill University 3-minute thesis competition, Guest Judge	*Institutional level*
2023	University of Paris – Cité M.Sc. in Informatics, "Digital Sciences" Program, Guest student member of the admission committee	*Institutional level*
2023	University of Paris – Cité M.Sc. in Informatics, "Digital Sciences" Program, Top 3 cGPA in the Class of 2023	*Institutional level*
2023	University of Paris – Cité, Summa Cum Laude [M.Sc. cGPA 4.0/4.0]	*Institutional level*
2021	University of Paris – Cité, Magna Cum Laude [M.Sc. cGPA 3.7/4.0]	*Institutional level*

Grant Writing

Independently developed and prepared the entire content of the following grant proposal:

2026	Early Professionals, Inspired Careers in AgeTech (EPIC-AT) CIHR subgrant proposal *"What Do We Measure? Advancing the Clinical Interpretability of Wearable Mobility Biomarkers in Older Canadians with Parkinson's Disease"*; Mihailidis A. (PI, University of Toronto), Paquette C. (Co-PI, McGill University), Mvomo C.E. (Applicant, McGill University); **[Awarded, CA$8,000 (for translating doctoral research findings into a clinical decision support system)]**	*Federal level*

Professional Experience

Research Experience

2023-Now *Montreal, Canada* | **McGill HBCL & Lake Lucerne Institute DART Groups,** Graduate Student Researcher
Advisors: Caroline Paquette and Chris Awai

Initiated a multi-award-winning Swiss–Canadian collaboration integrating neuroimaging, neuromotor control, AI, and wearable sensing to advance digital gait biomarkers in Parkinson's disease.

- Lead project conceptualization, funding acquisition, multimodal data curation and analysis, and dissemination through scientific publications and knowledge translation.
- Mentor undergraduate and master's students and support fellow PhD students in study conceptualization, data collection, and analysis.
- Contribute to the broader academic community through journal and conference peer review, academic committees, and grant evaluation and writing.

2023 — *Lausanne, Switzerland*

EPFL Rehabilitation & Assistive Robotics Group, Research Intern
Advisor: Alireza Manzoori

- Conducted multimodal analysis of inertial motion capture, exoskeleton encoder, force plate, and FSR data to identify the biomechanical factors explaining differences in assistance effectiveness.
- Contributed to experimental design and indoor and real-world multi-terrain data collection (e.g., inertial, metabolic, heart rate, …) to evaluate functional outcomes with and without assistance.

2022-2023 — *Paris, France*

French National Institute for Health & Medical Research (UMR 1284), Research Intern
Advisor: Kevin Lhoste

- Prototyped a mobile health application (Flutter SDK) for fall-risk assessment in older adults, based on simple balance tests using mobile sensing and machine learning.

2022 — *Vitznau, Switzerland*

Lake Lucerne Institute (LLUI) DART Group, Research Intern
Advisor: Chris Awai
Contributed to a multicenter trial integrating motion capture, virtual reality, and wearable sensing to improve balance and prevent falls in older adults.

- Led the LLUI trial arm, including participant recruitment, multimodal data collection, preliminary analysis, and dissemination to clinical and international audiences.
- Contributed to study conceptualization and collaboration with academic and industry partners.

2021 — *Compiègne, France*

UTC Biomechanics & BioEngineering (BMBI) Laboratory, Research Initiation Intern
Advisor: Khalil Ben Mansour

- Extracurricular internship during undergraduate studies. Conducted a mini-project on the effect of walking speed on peak plantar pressure. Tasks included data acquisition, basic signal processing, interpretation of biomechanical outputs, and results dissemination to lab members.

2021 — *Garches, France*

Greater Paris University Hospitals – Garches Foundation, Research Initiation Intern
Advisor: Anthony Supiot

- Extracurricular internship during undergraduate studies. Initiated to motion capture in neurorehabilitation. Assisted researchers during quantified gait analysis sessions.

Clinical Experience

2021-2023 — *Évry, France*

Simone Veil Community Health Center, Podiatrist Assistant
Referee: Prisca Lonn (Head Podiatrist)
Clinical practice (20%) in podiatric medicine conducted in parallel with the master's degree, with a focus on sports-related conditions, aging populations, and diabetic neuropathy.

- Activities included anamnesis, mobility assessment (e.g., gait and balance), design of custom medical devices (passive orthoses and prostheses), pharmacological and nonpharmacological treatment prescription, prevention and monitoring of complications, and communication with other healthcare professionals to support coordinated care.

2020 *Paris, France* | **IMSS Stade Français Paris Rugby Club,** Clinical Training – Sport Podiatric Medicine
Referee: Christophe Blanc (Head Podiatrist)

- Clinical training in elite sport (private sector). Assisted a senior sport podiatrist in anamnesis, mobility assessment, design of custom medical devices (passive orthoses), nonpharmacological treatment prescription, prevention and monitoring of complications, and communication with team physicians and therapists.

2018-2021 *Paris, France* | **Greater Paris University Hospitals (AP-HP),** Clinical Training – General Podiatric Medicine
Advisor: Yves Matillon (President of AFREP Institute at University of Paris – Cité)

- Clinical training within Europe's largest university hospital center. Hospital services included Internal Medicine, Dermatology, Rheumatology, Endocrinology, Diabetology, Rehabilitation Medicine, and Geriatrics.

Teaching Experience

2024-Now *Montreal, Canada* | **McGill Office of Science Education,** Science Education Fellow
Referee: Armin Yazdani (SciLearn Program Co-Lead)
Fellow with the McGill SciLearn team, a fellowship program preparing the next generation of science instructors.

- Managed "SciLearn Peer-Collaboration," a collaborative learning space supporting students in 10+ science courses. Develop and teach evidence-based neuroeducation content to 1,000+ undergraduate students and instructors in collaboration with cognitive neuroscientists and education developers. Conduct research on neuroeducation and organized student- and instructor-focused events.
- Co-founded "Étudions Ensemble," a university-wide program supporting francophone students, and collaborated with McGill administration to support its development.

2024-Now *Montreal, Canada* | **McGill Department of Kinesiology and Physical Education,** Teaching Assistant
Referee: Michael Creamer (Undergraduate Program Director)

- Assisted the lead instructor in the "Motor Control" course (B.Sc. Kinesiology) by addressing students' questions and difficulties with course material and assignments during office hours, mentoring students during tutorials, contributing to course and exam design, and supervising and grading written and oral examinations.

2024-2026 *Remote* | **Department of Rehabilitation, AFREP, University of Paris – Cité,** Lecturer
Referee: Yves Matillon (President of AFREP Institute at University of Paris – Cité)

- Developed and delivered 5×20 hours of online courses on the basics of research methodology (Bachelor of Podiatric Medicine). Activities included full course development, as well as coordination of guest lecturers.

Journal Articles

*Equal Contribution; Mentee

In Preparation

[J10] **C.E. Mvomo**, T. Ogawa, C. Guedes, and C. Paquette, "Quantifying Step Length During Treadmill Walking in Parkinson's Disease Using Inertial Sensors: A Validation Study," 2026

[J09] D. Leibovich, **C.E. Mvomo**, L. Higuita, J. Wu, C. Truong, I. Nunes Loureiro, A. Potvin-Desrochers, and C. Paquette, "Freezing of Gait in Parkinson's Disease: Does the New Definition Detect More Freezing?," 2026 **[Preliminary results received 3rd Price Best Undergraduate Oral Presentation at Minds in Motion Conference 2026]**

[J08] J.S.N Bedime, **C.E. Mvomo**, S. Perfetto, F. Parent-L'Ecuyer, H. Lajeunesse, I. Sierra, C. Guedes, JP Soucy, A. Potvin-Desrochers, M. Sharp, A. Rochette, A. Parent, and C. Paquette, "Brain Structural and Gait-Related Metabolic Correlates of Fatigue in Parkinson's Disease," 2026

[J07] **C.E. Mvomo**, S. Neumann, J. Pohl, M. Elgendi, and C.E. Awai, "Beyond Garbage In, Garbage

Out: Can Uncertainty Become Insightful in Real-World Outcome Prediction for Parkinson's Disease?," 2026

[J06] **C.E. Mvomo***, J.S.N. Bedime*, D.C. Hinton, T. Ogawa, A. Thiel, JP. Soucy, A. Potvin-Desrochers, and C. Paquette, "The Brain Network that Preserves Gait Adjustments in Parkinson's Disease," 2026

Under Review or In Revision *(with impact factors [IFs] at submission)*

[J05] C. Delaney, **C.E. Mvomo**, V. Gibson, C.E. Awai, P. Chitnis, and Q. Sanders, "Assistive Technologies, Clinical Considerations, and Future Directions for Managing Freezing of Gait in Parkinson's Disease," *Clinical Parkinsonism and Related Disorders*, under review, 2026 (IF: 2.7)

[J04] A. Parent, S. Perfetto, J.S.N. Bedime, F. Parent-L'Ecuyer, I. Sierra, H. Lajeunesse, **C.E. Mvomo**, M. Sharp, A. Rochette, A. Potvin-Desrochers, and C. Paquette, "The multidimensionality of fatigue in Parkinson's disease: a qualitative study of triggers, experience and coping strategies," *Disability and Rehabilitation*, under review, 2026 (IF: 2.2)

[J03] **C.E. Mvomo**, J.S.N. Bedime, T. Mitchell, A. Potvin-Desrochers, and C. Paquette, "A Multivariate Analysis of Frontoparietal Responses to Complex Walking in Freezing of Gait," *European Journal of Neuroscience*, in revision, 2026 (IF: 2.5)

[J02] **C.E. Mvomo**, J.S.N. Bedime, D. Leibovich, C. Guedes, A. Potvin-Desrochers, P.C. Dixon, C.E. Awai, and C. Paquette, "Gait-Related Digital Mobility Outcomes in Parkinson's Disease: New Insights into Convergent Validity?," *IEEE Transactions on Biomedical Engineering (TBME)*, in revision, 2026 (IF: 4.5) **[invited contribution as one of the top submission at BHI 2025]**

Published or In Press *(with impact factors [IFs] at submission)*

[J01] S. Neumann, **C.E. Mvomo**, D. Ravi, F. Schulte, L. Assländer*, and C.E. Awai*, "Visual Dependence for Postural Control is Increased in Older Adults," *Aging and Disease*, 2025 (IF: 9.6)

Conference Proceedings

All proceedings are heavily peer-reviewed and either published or in press; Mentee

[C06] **C.E. Mvomo**, M. Elgendi, and C.E. Awai, "Epistemic Uncertainty as a Driver of Performance in Wearable Monitoring of Parkinson's Disease: A Real-World Proof-of-Concept Study," *in Converging Clinical and Engineering Research on Neurorehabilitation – Proceedings of the 7th International Conference on NeuroRehabilitation (ICNR)*, in press, 2026

[C05] J.Y. Park, E. Morales Zaldivar, J.S.N. Bedime, **C.E. Mvomo**, F. Parent-L'Ecuyer, I. Sierra, H. Lajeunesse, A. Parent, M. Sharp, A. Rochette, and C. Paquette, "Examining the Impact of Fatigue on White Matter tracts in Parkinson's Disease," *in Movement Disorders*, in press, 2026

[C04] T. Ogawa, C. Guedes, **C.E. Mvomo**, J.S.N. Bedime, L. Hajjaji, A. Potvin-Desrochers, J. Choi, and C. Paquette, "Can people with Parkinson's disease improve gait adaptation skills via upregulation of the posterior parietal cortex?," *in Movement Disorders*, in press, 2026

[C03] **C.E. Mvomo**, J.S.N. Bedime, S. Perfetto, D. Leibovich, C. Guedes, A. Potvin-Desrochers, P.C. Dixon, C.E. Awai, and C. Paquette, "Monitoring Parkinson's Disease In-the-Wild," *in Proceedings of the IEEE-EMBS International Conference on Biomedical and Health Informatics*, 2025 **[500+ submissions, Acceptance rate <30%, 3rd Prize Best Paper Award, Top 0.6%]**

[C02] I. Sierra, S. Perfetto, A. Potvin-Desrochers, **C.E. Mvomo**, J.S.N. Bedime, H. Lajeunesse, F. Parent-L'Ecuyer, and C. Paquette. "Influence of Fatigue in Parkinson's Disease on Variability of Gait Measures," *in Movement Disorders*, 2024

[C01] **C.E. Mvomo**, J.S.N. Bedime, I. Sierra, S. Perfetto, H. Lajeunesse, A. Potvin-Desrochers, C.E. Awai, and C. Paquette, "Model-Based Gait Outcomes to Track Parkinson's Disease Progression?," *in Movement Disorders*, 2024

Selected Conference Posters

All peer-reviewed and limited to work not published in conference proceedings or presented orally; Mentee

[P08] J.S.N. Bedime, **C.E. Mvomo**, S. Perfetto, I. Sierra, H. Lajeunesse, F. Parent-L'Ecuyer, T. Ogawa, J.Y. Park, M. Sharp, MH. Pennestri, A. Potvin-Desrochers, B.G.G. da Costa, A. Parent, and C. Paquette, "Physical Activity and Sedentary Behavior Monitoring in People with Parkinson's Disease and Fatigue," *in International Society of Posture and Gait Research (ISPGR) - World Congress*, (Maastricht, Netherlands), 2025

[P07] J.S.N. Bedime, **C.E. Mvomo**, S. Perfetto, I. Sierra, H. Lajeunesse, JP. Soucy, A. Potvin-Desrochers, M. Sharp, A. Rochette, A. Parent, and C. Paquette, "Fatigue in Parkinson's Disease Is Linked to Enhanced Activity of the Fornix and Dysregulation of the Dorsal Attention and Central Executive Networks During Complex Walking: An [18F]-FDG PET Study," *in Annual Canadian Movement Disorders Meeting*, (Toronto, Canada), 2024

[P06] S. Perfetto, A. Potvin-Desrochers, I. Sierra, **C.E. Mvomo**, K. Shoemaker, M. Chakravarty, and C. Paquette, "Long-term, high-level aerobic training and the age-related decline in cortical thickness," *in Society for Neurosciences (SfN) World Congress*, (Chicago, USA), 2024

[P05] **C.E. Mvomo**, J.S.N. Bedime, S. Perfetto, H. Lajeunesse, I. Sierra, A. Potvin-Desrochers, and C. Paquette, "A Gait Model That Reflects Measures of Symptom Severity in Parkinson's Disease," *in 5th Quebec Congress in Adaptation-Rehabilitation Research*, (Quebec City, Canada), 2024

[P04] **C.E. Mvomo**, J.S.N. Bedime, S. Perfetto, H. Lajeunesse, I. Sierra, A. Potvin-Desrochers, and C. Paquette, "A Gait Model That Reflects Measures of Symptom Severity in Parkinson's Disease: A Pilot Study," *in Minds in Motion Conference*, (Montreal, Canada), 2024

[P03] J.S.N. Bedime, **C.E. Mvomo**, S. Perfetto, H. Lajeunesse, I. Sierra, A. Potvin-Desrochers, and C. Paquette, "The Influence of Cognitive Load on Locomotor Adaptation to Split-Belt Treadmill Walking in Parkinson's Disease with Freezing of Gait," *in Minds in Motion Conference*, (Montreal, Canada), 2024

[P02] S. Neumann, **C.E. Mvomo**, D. Ravi, F. Schulte, L. Assländer, and C.E. Awai, "Relative Importance of the Visual System for Balance Control Is Reflected in the Continuous Margins of Stability While Walking," *in XXIX Congress of International Society of Biomechanics (ISB)*, (Fukuoka, Japan), 2023

[P01] **C.E. Mvomo**, D. Ravi, F. Schulte, L. Assländer, and C.E. Awai, "Does Locomotor Perturbation Training Induce a Change in Relative Sensory Contribution to Balance? A Pilot Study," *in International Neurorehabilitation Symposium (RehabWeek)*, (Rotterdam, Netherlands), 2022

Intellectual Property

[IP02] **C.E. Mvomo**, "GaitHub: A digital health companion designed to democratize digital mobility biomarker analytics." Software, © 2026, MIT License.

[IP01] **C.E. Mvomo**, "PyComplexity: A Python module for computing the Attractor Complexity Index (ACI), a non-linear measure of gait automaticity." Software, © 2025, MIT License.

Oral Presentations

I delivered all oral presentations listed below

Primary Research Presentations

2025 | IEEE-EMBS Conference on Biomedical and Health Informatics (BHI), Atlanta, USA
"Monitoring Parkinson's Disease In-the-Wild" **[Selected as Top 9.2% best submissions]**

Knowledge Translation Presentations

2024 | TED-like Talk at NeuroLingo, Montreal, Canada **[10K+ views on YouTube]**

	"What if walking could help track the progression of Parkinson's disease?"
2024	CRIR 2024 Student Colloquium, Montreal, Canada **[People's choice award]**
	"What if walking became the key to monitoring the progression of Parkinson's disease?"

Science Education Research Presentations

2026	Supporting Active Learning (SALTISE) Conference, Montreal, Canada "Exploring the Link Between Belief in Neuromyths and Academic Performance in Undergraduate Science Students"
2025	Skills for Success Symposium, Montreal, Canada "McGill SciLearn Program" *(with Kira Smith)*

Invited Conference Presentations

2026	AGE-WELL 11th Annual Conference, Ottawa, Canada "Wearable Clinical Decision Support Systems for Older Adults with Parkinson's Disease: A Case Study of Decision Deferral Under Uncertainty"
2026	Nature Conferences "Redefining Healthcare in the Age of AI", Paris, France "When AI Defers: Uncertainty-Aware Wearable AI for Clinical Decision Support in Parkinson's Disease"
2025	IEEE-EMBS Conference on Biomedical and Health Informatics (BHI), Atlanta, USA "NextGen Scholar Presentation: Cyrille E. Mvomo"

Teaching

Graduate Student Instructor (Bachelor of Podiatric Medicine)

Winter 2026	Introduction to Research Methods – 1 (U.E. 5.3, ~ 40 students)	University of Paris – Cité
Fall 2025	Introduction to Research Methods – 2 (U.E. 5.4, ~ 20 students)	University of Paris – Cité
Winter 2025	Introduction to Research Methods – 1 (U.E. 5.3, ~ 20 students)	University of Paris – Cité
Fall 2024	Introduction to Research Methods – 2 (U.E. 5.4, ~ 35 students)	University of Paris – Cité
Winter 2024	Introduction to Research Methods – 1 (U.E. 5.3, ~ 35 students)	University of Paris – Cité

Teaching Assistant (B.Sc. Kinesiology)

Fall 2026	Motor Control (EDKP 447, ~ 80 students)	McGill University
Fall 2025	Motor Control (EDKP 447, ~ 80 students)	McGill University
Fall 2024	Motor Control (EDKP 447, ~ 70 students) *(with Gleydciane Fernandes)*	McGill University

SciLearn Guest Lectures (in large undergraduate science courses, each from 100 to 500+ students)

Developed and delivered lectures (individually or collaboratively) on "Learning how to learn through neuroscience"

Fall 2026	Introduction to Psychological Statistics (PSYC 204)	McGill University
Fall 2026	Introductory Organic Chemistry 1 (CHEM 212)	McGill University
Fall 2026	Algorithms and Data Structures (COMP 251)	McGill University
Winter 2026	Foundations of Programming (COMP 202) *(with Jean-Christophe Boivin)*	McGill University
Winter 2026	Introductory Organic Chemistry 2 (CHEM 222)	McGill University
Fall 2025	General Chemistry 1 (CHEM 110)	McGill University
Winter 2025	General Chemistry 2 (CHEM 120) *(with Hilary Sweatman)*	McGill University
Winter 2025	Introduction to Psychological Statistics (PSYC 204) *(with Hilary Sweatman)*	McGill University

Mentoring

Mentored 14 research projects, leading to conference and journal authorships (see publications/posters at pages 4-6).

Graduate Students

2025-Now	Clara Guedes, M.Sc. Student, McGill University **[2026 Canada CIHR Master Research Scholarship Recipient]** Thesis: “Split-belt Treadmill Training to Improve Gait in Parkinson’s Disease” Next: Medical Student at Université de Montréal	Biomechanics & Neuroscience
2024-Now	Jae Young Park, M.Sc. Student, McGill University Thesis: “Fatigue-Related White Matter Changes in Parkinson’s Disease”	Biomechanics & Neuroscience
2024-Now	Tomoha Ogawa, M.Sc. Student, McGill University Thesis: “Parietal Cortex Neuromodulation to Improve Gait in Parkinson”	Biomechanics & Neuroscience
2026	Florian Gourdin, Visiting M.Eng. Student, Polytech Marseille Project: “Parietal Cortex Upregulation to Improve Gait in Freezing of Gait” Next: Final Year M.Eng. Student at Polytech Marseille	Biomedical Engineering

Undergraduate Students

2026	Daniela Garcia Garcia, Visiting B.Sc. Student, Universidad de Monterrey **[2026 Mitacs Globalink Summer Research Internship Program]** Project: “Inertial Step Length Validation on a Treadmill in Parkinson” Next: Final Year B.Sc. Student at Universidad de Monterrey	Biomedical Engineering
2026	Marilena Sahlas-Roy, B.Sc. Student, McGill University Project: “Annotating Ziegler Test Recordings in Parkinson’s Disease” Next: 2nd Year B.Sc. Student, McGill University	Kinesiology
2026	Yasmine Abouelfareh, B.Sc. Student, McGill University Project: “Annotating Ziegler Test Recordings in Parkinson’s Disease” Next: 2nd Year B.Sc. Student, McGill University	Kinesiology
2026	Tatiana Amaya Quintero, B.Sc. Student, McGill University Project: “Annotating Ziegler Test Recordings in Parkinson’s Disease” Next: Final Year B.Sc. Student, McGill University	Kinesiology
2026	Justin Wu, B.Sc. Student, McGill University Project: “Annotating Freezing of Gait Recordings in Parkinson’s Disease” Next: 3rd Year B.Sc. Student, McGill University	Kinesiology
2026	Ines Nunes Loureiro, B.Sc. Student, McGill University Project: “Annotating Freezing of Gait Recordings in Parkinson’s Disease” Next: Final Year B.Sc. Student, McGill University	Kinesiology
2026	Ciara Truong, B.Sc. Student, McGill University Project: “Annotating Freezing of Gait Recordings in Parkinson’s Disease” Next: Senior Leader at Vancouver Champlain Heights Community Centre	Kinesiology
2024-2026	Dahlia Leibovich, B.Sc. Student (Honors), McGill University **[2026 Canada CIHR Master Research Scholarship Recipient]** Thesis: “Evaluating Freezing of Gait Severity in Parkinson’s Disease” Next: M.Sc. Student in Kinesiology at the University of Toronto	Kinesiology
2024	Laura Higuita, B.Sc. Student, Universidad Autónoma de Occidente **[2024 Mitacs Globalink Summer Research Internship Program]** Thesis: “Detecting Freezing of Gait in Parkinson’s Disease” Next: Final Year B.Sc. Student at Universidad Autónoma de Occidente	Biomedical Engineering

Media Coverage

2026	Funded Research – 2026 Research competition	Parkinson Canada
2026	SciLearn: Learning how to learn	The Tribune
2026	KPE PhD Candidate Cyrille Mvomo Honored with IEEE-EMBS BHI 2025 Best Paper Award and NSF-EMBS-Google NextGen Scholar Recognition	McGill News
2024	Results – CRIR 2024 Student Colloquium	CRIR News

Entrepreneurship

2026-Now | Co-founder and manager of "Étudions Ensemble," McGill's first university-wide program providing undergraduate students with a space to study in French and build connections. The program attracted students from across the university and secured support from McGill's administration and Office of Science Education.

Professional Service

Undergraduate Theses Defense Committee Member

2026 | Xavier Surget, AFREP Institute, University of Paris – Cité
Thesis: "Effect of Foot Orthoses on Pressure Points of a Flexible Flat Foot"

2026 | Flavie Tran Dinh, AFREP Institute, University of Paris – Cité
Thesis: "Podiatric Management of Blisters in Desert Environments"

2026 | Coline Vollant, AFREP Institute, University of Paris – Cité
Thesis: "Effect of Running on Foot Posture in Novice and Trained Runners"

2025 | Zoé Imbault, AFREP Institute, University of Paris – Cité
Thesis: "The Impact of Raynaud's Phenomenon on Skiers"

2025 | Rachelle Olivette, AFREP Institute, University of Paris – Cité
Thesis: "Impact of Foot Orthoses on Low Back Pain"

2025 | Hugo Forgez, AFREP Institute, University of Paris – Cité
Thesis: "Running Shoes : Prevention and Treatment of Plantar Fasciitis"

Conference Organizer

2026 | McGill Undergraduate Science Showcase, Member of the Organizing Team
Conference aimed at providing undergraduate science students with the opportunity to present their research.

2025 | McGill Undergraduate Science Showcase, Member of the Organizing Team
Conference aimed at providing undergraduate science students with the opportunity to present their research.

Workshop Organizer

Fall 2026 | SciLearn Workshop at Discover McGill, McGill University, Facilitator (with JC. Boivin)
Workshop aimed at providing undergraduate science students with basic knowledge of neuroeducation.

Winter 2026 | SciLearn Workshop "Prep for Finals," McGill University, Facilitator (with JC. Boivin)
Workshop aimed at providing undergraduate science students with basic knowledge of neuroeducation for the finals.

Winter 2026 | SciLearn Workshop "MidTerm Bounce-Back," McGill University, Facilitator (with JC. Boivin)
Workshop aimed at providing undergraduate science students with basic knowledge of neuroeducation after the mid-terms.

Fall 2025 | SciLearn Orientation Week Workshop, McGill University, Facilitator
Workshop aimed at providing undergraduate science students with basic knowledge of neuroeducation.

Fall 2025 | SciLearn Training Workshop "How Students Learn," McGill University, Facilitator (with A. Yazdani)
Workshop aimed at providing teaching assistants in Chemistry with basic knowledge of neuroeducation.

Winter | SciLearn Workshop "Prep for Finals," McGill University, Facilitator (with H. Sweatman)

2025 *Workshop aimed at providing undergraduate science students with basic knowledge of neuroeducation for the finals.*

Winter 2025 SciLearn Workshop "MidTerm Bounce-Back," McGill University, Facilitator (with H. Sweatman)
Workshop aimed at providing undergraduate science students with basic knowledge of neuroeducation after the mid-terms.

Fall 2024 SciLearn Workshop "Prep for Finals," McGill University, Facilitator (with H. Sweatman)
Workshop aimed at providing undergraduate science students with basic knowledge of neuroeducation for the finals.

Fall 2024 SciLearn Workshop "MidTerm Bounce-Back," McGill University, Facilitator (with H. Sweatman)
Workshop aimed at providing undergraduate science students with basic knowledge of neuroeducation after the mid-terms.

Conference Reviewer

2026 IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI'26)
Review of 5 articles on Precision Health and AI in Parkinson.

2025 IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI'25)
Review of 6 articles on Precision Health and AI. **[Reviewer Recognition]**

Journal Reviewer

2024 Medicine & Science in Sports & Exercise
Ad hoc review (via supervisor) of an article on wearable technologies for monitoring mobility disorders in Parkinson.

Academic Committee Member

2024-Now Committee on Innovative Practices in Research Ethics in Rehabilitation (PIER), CRIR
Student member of a committee that supports CRIR research teams in addressing emerging research ethics challenges, such as participant recruitment, digital data management, and support tools.

2023-Now Research Committee, Rehabilitation Department, University of Paris – Cité
Alternate member of a committee dedicated to promoting research within the newly created department.

Social Mobilization

2024 Patient Engagement Workshop, Parkinson Canada
Participation in a workshop aimed at improving interaction between researchers, patients, clinicians and industrial partners to enhance patient engagement in Parkinson's research in Canada.

2022 Hackathon, HackaHealth Zürich
Participated in hackathon to develop solutions for people with disabilities.

2022 Fundraising Events, Or Bleu
Participation in soccer tournaments to finance the construction of water wells in Mali.

Professional Development

2026-Now EPIC-AT Training
Interdisciplinary training in AgeTech innovation to translate research findings into digital solutions for older Canadians.

2024-Now McGill Science Education Fellow Program
Training to become next-generation science faculty through lectures/workshops on science education and faculty mentoring.

2024 V1 Studio Researchers and Entrepreneurs Network Program (RCE)
Entrepreneurship training for researchers to translate research findings into real-world solutions.

2023 CIHR Integrating Sex and Gender into Health Research
Training in integrating sex and gender considerations into health research design, methods, analysis, and interpretation.

2023 Government of Canada TCPS 2 CORE-2022 (Research Ethics)
Training in the ethical conduct of research involving humans under Canada's Tri-Council Policy Statement (TCPS 2).

2022 Motek Medical Gait Analysis & Rehabilitation using CAREN (Operator Level 1)
Training on fully immersive multimodal motion analysis (e.g., optical, inertial, force plates, EMG, ...) using CAREN.

2021 French Ministry of Health AFGSU 1 and 2 (emergency care)
Training in first aid for healthcare professionals, including recognition and initial management of medical emergencies.

Professional Affiliations & Licensure

2025-Now	Student Member, Institute of Electrical and Electronics Engineers (IEEE)
2025-Now	Student Member, IEEE Engineering in Medicine and Biology Society (EMBS)
2024-Now	Student Member, Movement Disorders Society (MDS)
2023-Now	Student Member, Montreal Centre for Interdisciplinary Research in Rehabilitation (CRIR)
2023-Now	Student Member, CRBLM, Faculty of Medicine and Health Sciences, McGill University
2023-2026	Student Member, Quebec BioImaging Network (QBIN)
2021-2023	Member, French Podiatric Medicine Society (SOFPOD)
2021-2023	Registered Podiatrist, French National Council of Podiatrists (ONPP)

References

Dr. Caroline Paquette
McGill University
475 Pine Avenue West Montreal, Quebec H2W 1S4
caroline.paquette@mcgill.ca

Dr. Chris Easthope Awai
Lake Lucerne Institute AG
Rubistrasse 9 6354 Vitznau
chris.awai@llui.org

Dr. Philippe C. Dixon
McGill University
475 Pine Avenue West Montreal, Quebec H2W 1S4
phil.dixon@mcgill.ca

Dr. Armin Yazdani
McGill University
2001 McGill College Montreal, Quebec H3A 1G1
armin.yazdani@mcgill.ca